

Google Panel 2022: Peering into the Work World of Computer Scientists

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On December 13th, were you in Murray Kahn Theater to learn about some of the inner workings of the Google offices just a subway ride away on 8th Avenue? Thanks to Mr. Dillon, computer science teacher in his first year at Stuy, and Ms. Prabhu, Assistant Principal of Biology, Computer Science and Special Education, a panel of four guest speakers gave Stuy students and faculty members an insightful glimpse into some different roads and different roles within the tech giant. Tamiyyah Shafiq and Alexander Panas, sophomores from Mr. Dillon’s CS class, served as welcoming hosts for this event by providing opening remarks, introducing the panelists, and coordinating the Q&A.

1 Why Host a Google Panel, and Why Now?

In an email interview, Mr. Dillon described his motivation for initiating this opportunity, saying, “Computer Science Education Week (CSEdWeek) is an annual event during which teachers engage students to learn computer science. Schools around the country host special events like hack-a-thons, Hour of Code, or guest talks with people in tech. Industry partners often get involved, which is where Google came in. My motivation... was to share the guest speakers’ stories about their path to a career in tech. I’ve always been a teacher, so I don’t have a personal story about finding a job in a tech company like Google.” He credited Ms. Prabhu for her crucial help with a wide variety of logistics, as well as a Google Program Manager, Mr. Dion Paul, for serving as liaison and providing the panel members who generously volunteered their time and their stories.

So who were these informative Googlers who shared their backgrounds and their industry perspectives with us? **Mr. Andrew Myers**, a network architect with Google Distributed Cloud

Hosted, is a “Noogler,” having worked for Google for nine months, designing networking systems that can quickly transfer data reliably and efficiently. He earned an M.S. in Applied Mathematics from Odessa University and an M.S. in Management from Harvard University. **Ms. Catherine Liang**, a software engineer with the Android System User Interface team, also a “Noogler,” works with designers to implement design and functionality changes to the Android OS. She graduated last May from the University of Pennsylvania with a B.S. in Computer Science and Computer Graphics. **Mr. Benjamin Taubenblatt**, has worked with Google for two years as a software engineer on its Speech Recognition Infrastructure. He builds systems that support speech-to-text technology for products like the Google Assistant. He graduated from McGill University with a B.S. in Computer Science and Linguistics in 2019. **Dr. Pratik Worah**, a senior research scientist with Google Research, has worked at Google for about eight years on probability theory, neural networks, and artificial intelligence underlying algorithms. He earned a Ph.D. in Theoretical Computer Science from the University of Chicago in 2013, following his undergraduate studies in Computer Science and Engineering at Indian Institute of Technology, Kharagpur and his M.S. in Applied Mathematics at University of Illinois, Urbana.

2 Googlers and Their Stories

Each Google panelist took the mic to offer their individual career paths, starting with Ben Taubenblatt. Surprisingly, he initially had a negative experience with math in high school, nearly dropping AP Calculus in senior year. “I was never someone who was super good at math, never studied the right way for it,” he recalled. But he changed how he approached the material and slowly improved, ultimately earning that AP credit, which inspired him to take higher level math and computer science classes in college and then to major in computer science. During college, he completed software engineering internships at FINRA and Morgan Stanley. After graduation, he worked in consulting at Deloitte, in its Big Data and AI department. After one year, at the start of the pandemic, he was assigned to business intelligence projects, but quickly realized that he was passionate about software engineering and programming, so he quit after just two weeks. Eventually, a friend referred him to job opportunities at Google, and he embarked on the lengthy application process that entailed hard programming/brain teaser interview problems.

Like many teams at Google, he codes primarily in C++. He builds systems for speech researchers at Google who host machine learning models, the core technology behind the conversion of a raw audio query into text. Specifically, his team handles the ingestion of large amounts of speech data for those models, the transcription and annotation of that data, and checking in data sets for training the models.

Next, Catherine Liang described how she began her “great experience” at Google. In college, she studied digital media design, including e-design and automation. As a high school student in New Jersey, she was not exposed to much in the way of computer science education and was more interested in art. To be practical, she decided to pursue college studies in data graphics to open up tech job prospects while still utilizing her skills in art. She started as a Deloitte tech consultant, working on infrastructure and setting up JavaScript frameworks.

When a Google recruiter reached out to her after her first year at Deloitte, she was already interested in applying, since she had always wanted to work for Google in college. She went through three rounds of technical interviews and one round of behavioral before she was matched with the

Android OS UI team. Using Android Studio, Java, and Kotlin, she contributes to various aspects of the frontend user experience. To share a fun fact with us, she mentioned that she created games using Unity and Unreal Engine!

Andrew Myers came prepared to show off his brightly-colored Noogler hat, complete with a small plastic propeller on the top! He is still learning about Google culture in his first year by asking a lot of questions. Despite his significant experience in systems at large companies such as Morgan Stanley, Microsoft, and AT&T, his journey to Google employment was “not quick,” receiving an offer only after persistently applying three separate times (five years ago, two years ago, and less than one year ago). He emphasized that foundations in both math and computer science are equally important to be successful in tech.

In his team’s work designing networks, he is building the foundation of Google Cloud, and he uses the platform Anthos extensively. One important aspect of network design is security, including protection from both internal and external attacks, as well as compliance with government requirements. He offered this accomplishment as a fun fact about him: He once rebooted just one router to restore Internet service to the whole tri-state area!

Pratik Worah shared that he grew up in India and was originally “crazy about physics.” He admitted that when he first studied computer science, he was “right at the middle” of his class, while his classmates were much more familiar with programming in C. But he continued with his math and computer science studies on “more or less autopilot” until he completed his Ph.D. in Theoretical Computer Science. Much of his research focused on complexity theory, which involves proving impossibility, one of the deepest areas of mathematics.

After a little over one year of post-doc work at NYU’s Courant Institute of Mathematical Sciences, he spent slightly over one year at a global finance firm, PIMCO (Pacific Investment Management Company), where his research involved mathematical models for bond and derivative investment strategies. He has enjoyed working on a diverse portfolio of problems, some involving economics or game theory, and commented on the applicability of calculus to find maxima and minima. Although he had worked briefly for Google in Mountain View as a software engineer in between getting his Master’s degree and his Ph.D., it was not until his colleague encouraged him to work for Google Research that he left his job as a quantitative researcher at PIMCO and began his work as a Google Senior Research Scientist, focusing on neural networks and their limitations.

3 We Had Questions, They Had Answers

Attendees kept the discussion going with their questions for the panelists, submitted via QR code. When asked for advice about how to get **internships**, Ben Taubenblatt remarked that internships provided him with his first taste of real, “out in the wild” programming at a company, where he saw how teams and companies operate. He acknowledged that getting the first internship is the hardest, since you don’t have any work experience on your resume. He encouraged students to make their resumes stand out by doing personal projects, which show passion. For example, if you can build a simple webpage, both the front end and the back end, and include a URL on your resume that a recruiter can click on. He advised that your resume should be concise, one page, since recruiters at large companies only spend 10 to 20 seconds on each one, and you want the most important items to stand out.

Catherine Liang reflected that she had a different experience than many of her peers, since she didn't intern at any big tech companies, like most software engineers. She advised that it's good to start interning early, noting that some of her classmates interned during the summer after freshman or sophomore year, and were able to get return offers for the summer after junior year, leading to a full-time offer after graduation. She also encouraged students to try their best when they apply for internships, even though the process is highly competitive. But she noted that if an internship opportunity doesn't work out, she hopes students "get more excited about doing your own projects." For example, she gained software development experience by working on fun game design with friends over the summer.

Andrew Myers emphasized the importance of starting to build a network "from the beginning," by getting to know people in tech companies. That way, when the time comes to get internships, your network can help you get a head start over others.

Students also inquired about the most **enjoyable/rewarding parts** of working in the panelists' different fields. Andrew Myers noted how network engineers can bring back lost foundational services and allow vital computer systems to withstand drastic events. Catherine Liang remarked that her software engineering work on frontend Android functionalities makes smartphone use "bug free" and more easy and enjoyable. Ben Taubenblatt shared that he loves how the coding process in general is "extremely creative" and that problem-solving does not just involve math and logic. He described how people approach the same high-level problems completely differently. He finds his work of developing solutions to support the machine learning models of speech recognition systems does not have a "defined playbook," so Google is a trailblazer on the leading edge of that technology.

Lastly, audience members asked panelists what they look forward to the most in the **future of computer science**. Pratik Worah predicted that computer science will become even more foundationally useful in more diverse ways but likened computer science to a screwdriver for a mechanic. He opined that computer science doesn't have to be the only focus of life, saying, "No one studies a screwdriver!"

Ben Taubenblatt looks forward to the future of quantum computing, "the next big jump." Just as transistors allow the smartphone in our pockets to be more powerful than the computers from the 1950s which took up entire rooms, he predicts that having the power of quantum computers in our pockets will be incredible.

4 Our Attitude of Gratitude

The Stuy CS community sincerely thanks these Googlers for their enlightening and enthusiastic information and advice. Mr. Dillon reflected on the panel discussion, saying, "I think the guest speakers did a great job of sharing stories of perseverance and hard work with our students. Many of my students who attended found those stories inspiring and took time after the event to meet the panelists."

To learn more about Computer Science Education Week and its "call to action," we encourage our readers to visit <https://www.csedweek.org/> and explore all the various resources and opportunities.